BiasAperture — Demographic Fairness Audit Dossier

Diagnostic Demographic Bias Evaluation & Regulatory Traceability Report

AUDITED MODEL
FairFace ResNet-34 Multi-Task Classifier

EVALUATION DATASET
FairFace Benchmark Dataset (N=10954)

PROTECTED AXIS
Race_gender

AUDIT DATE
2026-10-19 03:58:24 UTC

Diagnostic Scope & Non-Intervention Guarantee:

BiasAperture is engineered exclusively as an external diagnostic auditing, statistical significance testing, and regulatory traceability platform. It does **not** perform model retraining, in-processing debiasing, or synthetic image generation. In adherence to statistical integrity safeguards (NFR-003), demographic subgroups with fewer than 30 samples (*n < 30*) have point estimates suppressed to prevent unreliable compliance claims.

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SUBGROUPS WITH SIGNIFICANT DISPARITY (P < 0.05)

Overall (Table 1)
WORST DISPARITY COHORT (32 METRIC FAILURES)

DIR = 0.025
WORST HEADLINE METRIC (OVERALL)

0
SUBGROUPS FLAGGED N < 30 (NFR-003)

1. Headline Fairness Metrics (Core Four — Cross-Group Overview)

Summary metrics evaluated across all eligible demographic strata (n ≥ 30 per NFR-003). Accompanied by 95% stratified BCa bootstrap confidence intervals and Pearson's χ^2 test of independence. Plotted on a calibrated [0.0, 1.0] domain with null-point reference and threshold lines.

Metric Name	Point Estimate	95% Bootstrap CI	CI Band vs. Null	Significance (p-value)	Regulatory Clause
Demographic Parity Difference	0.955	[0.941, 0.961]		p=0.0000 **p < 0.05**	EU AI Act Art. 10(2)(f) / NIST Measure 2.11
Equalized Odds Difference	N/A (Cond. Support)	—	—	—	EU AI Act Art. 10(2)(g) / NIST Measure 2.11
Equal Opportunity Difference	0.053	[0.032, 0.067]		p=0.0000 **p < 0.05**	EU AI Act Art. 10(3) / NIST Measure 2.11
Disparate Impact Ratio	0.025	[0.019, 0.035]		p=0.0000 **p < 0.05**	EEOC 4/5ths Rule / EU AI Act Annex IV §2(g)

2. Per-Subgroup Disparity Matrix

Detailed stratification across individual demographic groups, ordered by largest deviation from the null (no-disparity) point, most significant first. Subgroups below the NFR-003 sample threshold (n < 30) are strictly flagged, uncomputed to prevent small-sample estimation artifacts, and summarised separately below.

Subgroup	Sample Size (n)	DPD	EOD (Max-Gap)	EOP	DIR (Symmetric)	Deviation from Null (red = p < 0.05)	Data Integrity Status
Indian_Female	763	0.500	—	—	0.047	DPD DIR	Audited (Valid)
Southeast Asian_Female	680	0.497	—	—	0.053	DPD DIR	Audited (Valid)
East Asian_Female	773	0.489	—	—	0.069	DPD DIR	Audited (Valid)
White_Female	963	0.489	—	—	0.069	DPD DIR	Audited (Valid)
Latino_Hispanic_Female	830	0.487	—	—	0.073	DPD DIR	Audited (Valid)
Black_Female	757	0.476	—	—	0.093	DPD DIR	Audited (Valid)
Middle Eastern_Female	396	0.472	—	—	0.101	DPD DIR	Audited (Valid)
White_Male	1122	0.454	—	0.019	0.536	DPD EOP DIR	Audited (Valid)
East Asian_Male	777	0.453	—	0.018	0.537	DPD EOP DIR	Audited (Valid)
Middle Eastern_Male	813	0.445	—	0.010	0.541	DPD EOP DIR	Audited (Valid)
Indian_Male	753	0.439	—	0.004	0.545	DPD EOP DIR	Audited (Valid)
Southeast Asian_Male	735	0.435	—	0.000	0.547	DPD EOP DIR	Audited (Valid)
Latino_Hispanic_Male	793	0.412	—	0.024	0.561	DPD EOP DIR	Audited (Valid)
Black_Male	799	0.401	—	0.034	0.567	DPD EOP DIR	Audited (Valid)

3. Targeted Disparity Explainability & Proxy Attributions

Demographic Surrogate Associations — Mean absolute contribution to surrogate log-odds

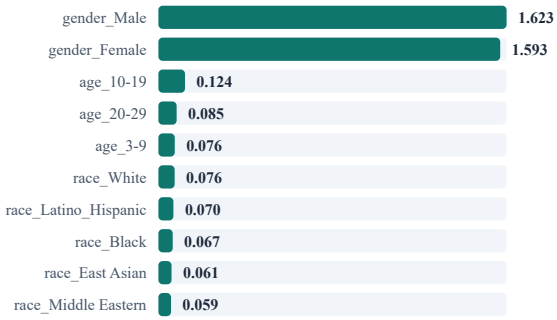
Surrogate additive Shapley attributions computed for statistically flagged disparities ($p < 0.05$, $n \geq 30$). A global ridge-regularized logistic surrogate model is fitted to predictions across the full evaluation dataset to estimate feature proxy influence. Mean absolute Shapley attributions measure the proxy contribution of each demographic feature to surrogate log-odds.

> **Non-Causal Disclaimer (Bilodeau et al., 2024):** Attributions identify empirical associations between demographic proxy features and model predictions. They do *not* establish causal relationships or prescribe remediation. This is a diagnostic tool only — no model retraining, in-situ weight modification, or causal intervention is performed.

Statistically Flagged Disparities Triggering Surrogate Analysis

ALL • DIR | ALL • DPD | ALL • EOP | Black_Female • DIR | Black_Female • DPD | Black_Male • DIR | Black_Male • DPD | Black_Male • EOP | East Asian_Female • DIR | East Asian_Female • DPD | East Asian_Male • DIR | East Asian_Male • DPD | East Asian_Male • EOP | Indian_Female • DIR | Indian_Female • DPD | Indian_Male • DIR | Indian_Male • DPD | Latino_Hispanic_Female • DIR | Latino_Hispanic_Female • DPD | Latino_Hispanic_Male • DIR | Latino_Hispanic_Male • DPD | Latino_Hispanic_Male • EOP | Middle Eastern_Female • DIR | Middle Eastern_Female • DPD | Middle Eastern_Male • DIR | Middle Eastern_Male • DPD | Southeast Asian_Female • DIR | Southeast Asian_Female • DPD | Southeast Asian_Male • DIR | Southeast Asian_Male • DPD | White_Female • DIR | White_Female • DPD | White_Male • DIR | White_Male • DPD | White_Male • EOP

Global Demographic Proxy Feature Importance



Demographic Feature	Mean absolute contribution to surrogate log-odds
gender_Male	1.6228
gender_Female	1.5932
age_10-19	0.1241
age_20-29	0.0850
age_3-9	0.0763
race_White	0.0758
race_Latino_Hispanic	0.0700
race_Black	0.0667

Demographic Feature	Mean absolute contribution to surrogate log-odds
race_East Asian	0.0612
race_Middle Eastern	0.0588

4. Governance Documentation & Intended Use

> **EU AI Act (Regulation EU 2024/1689) Article 10 / Annex IV Alignment:** This audit dossier provides verifiable diagnostic documentation of demographic performance disparities. It does not perform model retraining or in-situ weight modifications.

Model Description: ImageNet-pretrained ResNet-34 terminating in an 18-unit linear head sliced for race (7), gender (2), and age (9) classification.

Backend configuration: dual. Both backend executions completed; 0 point-estimate discrepancies detected.

Cross-backend agreement checks point estimates only; bootstrap, hypothesis tests, and reporting share implementation. Consult execution logs for backend failures and divergences.